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Leider; Colby Nelson

Voice user interface using non-linguistic input

Abstract

A voice user interface (VUI) and methods for operating the VUI are disclosed. In some embodiments, the VUI configured to receive and process linguistic and non-linguistic inputs. For example, the VUI receives an audio signal, and the VUI determines whether the audio input comprises a linguistic and/or a non-linguistic input. In accordance with a determination that the audio signal comprises a non-linguistic input, the VUI causes a system to perform an action associated with the non-linguistic input.

Inventors: Leider; Colby Nelson (Coral Gables, FL)
Applicant: MAGIC LEAP, INC. (Plantation, FL); Leider; Colby Nelson (Plantation, FL)
Family ID: 1000008819427
Assignee: Magic Leap, Inc. (Plantation, FL)
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Primary Examiner: Abebe; Daniel

Attorney, Agent or Firm: Morrison & Foerster LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This Application is a national stage application under 35 U.S.C. § 371 of International Application No. PCT/US2021/053046, filed internationally on Sep. 30, 2021, which claims the benefit of U.S. Provisional Application No. 63/085,462 filed on Sep. 30, 2020, the entire disclosure of which is herein incorporated by reference for all purposes.

FIELD

(1) This disclosure relates in general to systems and methods for operating a voice user interface configured to receive and process linguistic and non-linguistic inputs.

BACKGROUND

(2) Voice user interfaces (VUIs) may employ automatic speech recognition (ASR) (e.g., speech-to-text) coupled with a semantic model that maps spoken natural language (e.g., "please email Frank that I'll be late to the meeting") into intents and values (e.g., INTENT="SEND_EMAIL," RECIPIENT="Frank", BODY="I'll be late to the meeting"). Such utterances are often preceded by a Wake-Up Word (WuW), which instructs a speech system to wake from sleep and prepare to parse a user's utterance. This model may be used for systems such as home assistants, appliances, personal robots, and IoT devices, in which a substantially immediate or real-time response may not be critical.

(3) However, some scenarios may require a more immediate response and action (e.g., taking a photograph, recording a video, recording an audio). For example, VUI-enabled cameras and head-mounted augmented reality (AR) or mixed reality (MR) devices may need to be activated quickly to record a video or take a picture using a voice input, when activating such operation may not be convenient (e.g., a picture of a patient at a specific time need to be taken during surgery, a video of a MR environment need to be recorded during game play). In these cases, by the time a WuW is uttered and followed by a voice command (e.g., "take a picture") or a combination of inputs is entered on a device, too much time may have lapsed to capture a desired picture or recording. Furthermore, the time it takes to indicate certain actions may be unnecessarily long. This may also be the case, for example, with issuing graphical communication memes, such as emojis, which are increasingly prevalent in communication. Evoking an

emoji in a message may require one to either change the layout of a virtual keyboard and search for the desired emoji, or to remember and type a corresponding control sequence, which may be complicated, (e.g., “/smileyface” or “:-)” that is then mapped to an emoji character).

(4) In some scenarios, text-based communication (e.g., text, chat, email) using speech recognition may not convey subtleties of a sender's underlying emotion and/or intent. For example, if a sender sends a message intended to be a joke, the humor may be lost at the receiver's end because the words of the message themselves may not convey the sender's intentions.

(5) For these reasons, it would be desirable to improve the voice user interface to allow for quicker response and action on a device and convey a sender's underlying emotion and/or intent.

BRIEF SUMMARY

(6) A voice user interface (VUI) and methods for operating the VUI are disclosed. In some embodiments, the VUI is configured to receive and process linguistic and non-linguistic inputs. For example, the VUI receives an audio signal, and the VUI determines whether the audio input comprises a linguistic and/or a non-linguistic input. In accordance with a determination that the audio signal comprises a non-linguistic input, the VUI causes a system to perform an action associated with the non-linguistic input. For example, the non-linguistic input may be one of a paralinguistic input and a prosodic input. As an exemplary advantage, the VUI is able to respond to time-critical commands much closer to real-time than is possible with natural-language processing (NLP) systems.

(7) In some embodiments, a method comprises: receiving, using a microphone of a system, an audio signal; determining whether the audio signal comprises a non-linguistic input; and in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input.

(8) In some embodiments, the non-linguistic input is a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input.

(9) In some embodiments, the non-linguistic input is a prosodic input, and in accordance with a determination that the audio signal comprises the prosodic input, the action comprises a second action associated with the prosodic input.

(10) In some embodiments, the second action is a modification of an action associated with a linguistic input.

(11) In some embodiments, the prosodic input is indicative of an emotion associated with a user of the system, the user of the system associated with the audio signal, and the action is further associated with the emotion.

(12) In some embodiments, the method further comprises: determining whether the audio signal comprises a linguistic input; and in accordance with a determination that the audio signal comprises the linguistic input, performing a third action associated with the linguistic input.

(13) In some embodiments, the action comprises a modification of the third action based on the non-linguistic input.

(14) In some embodiments, the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, receiving information associated with the audio signal from a convolutional neural network, wherein the action is performed based on the information.

(15) In some embodiments, the method further comprises classifying a feature of the non-linguistic input, wherein the action is performed based on the classified feature.

(16) In some embodiments, the method further comprises associating the action with the non-linguistic input.

(17) In some embodiments, the action comprises one of texting, performing an intent, and inserting an emoji.

(18) In some embodiments, the method further comprises receiving, from a sensor of the system different from the microphone, information associated with an environment of the system, wherein the action is further associated with the information received from the sensor.

(19) In some embodiments, the system is a mixed reality system in a mixed reality environment, and the action is further associated with the mixed reality environment.

(20) In some embodiments, the method further comprises determining a position of the system, wherein the action is further associated with the position of the system.

(21) In some embodiments, in accordance with a determination that the system is associated with a first user, the action comprises a first action associated with the first user; and in accordance with a determination that the system is associated with a second user, different from the first user, the action comprises a second action associated with the second user, different from the first action.

(22) In some embodiments, the audio signal comprises a frequency-domain feature and a time-domain feature, and the determination of whether the audio comprises the non-linguistic input is based on the frequency-domain feature and the time-domain feature.

(23) In some embodiments, the method further comprises receiving information from a feature database, wherein the determination of whether the audio comprises the non-linguistic input is further based on the information.

(24) In some embodiments, determining whether the audio signal comprises the non-linguistic input comprises using a first processor, and the method further comprises in accordance with a determination that the audio signal

comprises the non-linguistic input, waking up a second processor to perform the action.

(25) In some embodiments, a system comprises: a microphone; and one or more processors configured to execute a method comprising: receiving, using the microphone, an audio signal; determining whether the audio signal comprises a non-linguistic input; and in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input.

(26) In some embodiments, the non-linguistic input is a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input.

(27) In some embodiments, the non-linguistic input is a prosodic input, and in accordance with a determination that the audio signal comprises the prosodic input, the action comprises a second action associated with the prosodic input.

(28) In some embodiments, the second action is a modification of an action associated with a linguistic input.

(29) In some embodiments, the prosodic input is indicative of an emotion associated with a user of the system, the user of the system associated with the audio signal, and the action is further associated with the emotion.

(30) In some embodiments, the method further comprises: determining whether the audio signal comprises a linguistic input; and in accordance with a determination that the audio signal comprises the linguistic input, performing a third action associated with the linguistic input.

(31) In some embodiments, the action comprises a modification of the third action based on the non-linguistic input.

(32) In some embodiments, the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, receiving information associated with the audio signal from a convolutional neural network, wherein the action is performed based on the information.

(33) In some embodiments, the method further comprises classifying a feature of the non-linguistic input, wherein the action is performed based on the classified feature.

(34) In some embodiments, the method further comprises associating the action with the non-linguistic input.

(35) In some embodiments, the action comprises one of texting, performing an intent, and inserting an emoji.

(36) In some embodiments, the system further comprises a sensor different from the microphone, wherein the method further comprises receiving, from the sensor, information associated with an environment of the system, wherein the action is further associated with the information received from the sensor.

(37) In some embodiments, the system is a mixed reality system in a mixed reality environment, and the action is further associated with the mixed reality environment.

(38) In some embodiments, the method further comprises determining a position of the system, wherein the action is further associated with the position of the system.

(39) In some embodiments, in accordance with a determination that the system is associated with a first user, the action comprises a first action associated with the first user; and in accordance with a determination that the system is associated with a second user, different from the first user, the action comprises a second action associated with the second user, different from the first action.

(40) In some embodiments, the audio signal comprises a frequency-domain feature and a time-domain feature, and the determination of whether the audio comprises the non-linguistic input is based on the frequency-domain feature and the time-domain feature.

(41) In some embodiments, the method further comprises receiving information from a feature database, wherein the determination of whether the audio comprises the non-linguistic input is further based on the information.

(42) In some embodiments, determining whether the audio signal comprises the non-linguistic input comprises using a first processor, and the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, waking up a second processor to perform the action.

(43) In some embodiments, a non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to execute a method comprising: receiving, using a microphone of a system, an audio signal; determining whether the audio signal comprises a non-linguistic input; and in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input.

(44) In some embodiments, the non-linguistic input is a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input.

(45) In some embodiments, the non-linguistic input is a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input.

(46) In some embodiments, the non-linguistic input is a prosodic input, and in accordance with a determination that the audio signal comprises the prosodic input, the action comprises a second action associated with the prosodic input.

- (47) In some embodiments, the second action is a modification of an action associated with a linguistic input.
- (48) In some embodiments, the prosodic input is indicative of an emotion associated with a user of the system, the user of the system associated with the audio signal, and the action is further associated with the emotion.
- (49) In some embodiments, the method further comprises: determining whether the audio signal comprises a linguistic input; and in accordance with a determination that the audio signal comprises the linguistic input, performing a third action associated with the linguistic input.
- (50) In some embodiments, the action comprises a modification of the third action based on the non-linguistic input.
- (51) In some embodiments, the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, receiving information associated with the audio signal from a convolutional neural network, wherein the action is performed based on the information.
- (52) In some embodiments, the method further comprises classifying a feature of the non-linguistic input, wherein the action is performed based on the classified feature.
- (53) In some embodiments, the method further comprises associating the action with the non-linguistic input.
- (54) In some embodiments, the action comprises one of texting, performing an intent, and inserting an emoji.
- (55) In some embodiments, the method further comprises receiving, from a sensor of the system different from the microphone, information associated with an environment of the system, wherein the action is further associated with the information received from the sensor.
- (56) In some embodiments, the system is a mixed reality system in a mixed reality environment, and the action is further associated with the mixed reality environment.
- (57) In some embodiments, the method further comprises determining a position of the system, wherein the action is further associated with the position of the system.
- (58) In some embodiments, in accordance with a determination that the system is associated with a first user, the action comprises a first action associated with the first user; and in accordance with a determination that the system is associated with a second user, different from the first user, the action comprises a second action associated with the second user, different from the first action.
- (59) In some embodiments, the audio signal comprises a frequency-domain feature and a time-domain feature, and the determination of whether the audio comprises the non-linguistic input is based on the frequency-domain feature and the time-domain feature.
- (60) In some embodiments, the method further comprises receiving information from a feature database, wherein the determination of whether the audio comprises the non-linguistic input is further based on the information.
- (61) In some embodiments, determining whether the audio signal comprises the non-linguistic input comprises using a first processor, and the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, waking up a second processor to perform the action.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A-1B illustrate exemplary wearable systems according to some embodiments of the disclosure.
- (2) FIG. 2 illustrates an exemplary handheld controller according to some embodiments of the disclosure.
- (3) FIG. 3 illustrates an exemplary auxiliary unit according to some embodiments of the disclosure.
- (4) FIGS. 4A-4B illustrate exemplary functional block diagrams of an exemplary wearable system according to some embodiments of the disclosure.
- (5) FIG. 5 illustrates an exemplary voice user interface system according to some embodiments of the disclosure.
- (6) FIG. 6 illustrates an exemplary voice input discriminator according to some embodiments of the disclosure.
- (7) FIG. 7 illustrates an exemplary voice user interface system according to some embodiments of the disclosure.
- (8) FIG. 8 illustrates an exemplary method of operating a voice user interface system according to some embodiments of the disclosure.

DETAILED DESCRIPTION

(9) In the following description of examples, reference is made to the accompanying drawings which form a part hereof, and in which it is shown by way of illustration specific examples that can be practiced. It is to be understood that other examples can be used and structural changes can be made without departing from the scope of the disclosed examples.

(10) FIG. 1A illustrates an example wearable head device **100A** configured to be worn on the head of a user. Wearable head device **100A** may be part of a broader wearable system that comprises one or more components, such as a head device (e.g., wearable head device **100A**), a handheld controller (e.g., handheld controller **200** described below), and/or an auxiliary unit (e.g., auxiliary unit **300** described below). In some examples, wearable head device **100A** can be used for virtual reality, augmented reality, or mixed reality systems or applications. Wearable head device **100A** can comprise one or more displays, such as displays **110A** and **110B** (which may comprise left and

right transmissive displays, and associated components for coupling light from the displays to the user's eyes, such as orthogonal pupil expansion (OPE) grating sets **112A/112B** and exit pupil expansion (EPE) grating sets **114A/114B**; left and right acoustic structures, such as speakers **120A** and **120B** (which may be mounted on temple arms **122A** and **122B**, and positioned adjacent to the user's left and right ears, respectively); one or more sensors such as infrared sensors, accelerometers, GPS units, inertial measurement units (IMUs, e.g. IMU **126**), acoustic sensors (e.g., microphones **150**); orthogonal coil electromagnetic receivers (e.g., receiver **127** shown mounted to the left temple arm **122A**); left and right cameras (e.g., depth (time-of-flight) cameras **130A** and **130B**) oriented away from the user; and left and right eye cameras oriented toward the user (e.g., for detecting the user's eye movements) (e.g., eye cameras **128A** and **128B**). However, wearable head device **100A** can incorporate any suitable display technology, and any suitable number, type, or combination of sensors or other components without departing from the scope of the invention. In some examples, wearable head device **100A** may incorporate one or more microphones **150** configured to detect audio signals generated by the user's voice; such microphones may be positioned adjacent to the user's mouth and/or on one or both sides of the user's head. In some examples, the one or more microphones **150** may be configured to receive a voice input or an audio signal (e.g., voice input **502**, voice input **602**, voice input **702**) for a voice user interface (e.g., VUI system **500**, VUI system **700**) of the wearable head device. In some examples, wearable head device **100A** may incorporate networking features (e.g., Wi-Fi capability) to communicate with other devices and systems, including other wearable systems. Wearable head device **100A** may further include components such as a battery, a processor, a memory, a storage unit, or various input devices (e.g., buttons, touchpads); or may be coupled to a handheld controller (e.g., handheld controller **200**) or an auxiliary unit (e.g., auxiliary unit **300**) that comprises one or more such components. In some examples, sensors may be configured to output a set of coordinates of the head-mounted unit relative to the user's environment, and may provide input to a processor performing a Simultaneous Localization and Mapping (SLAM) procedure and/or a visual odometry algorithm. In some examples, wearable head device **100A** may be coupled to a handheld controller **200**, and/or an auxiliary unit **300**, as described further below.

(11) FIG. **1B** illustrates an example wearable head device **100B** (that can correspond to wearable head device **100A**) configured to be worn on the head of a user. In some embodiments, wearable head device **100B** can include a multi-microphone configuration, including microphones **150A**, **150B**, **150C**, and **150D**. In some examples, the one or more microphones **150** may be configured to receive a voice input or an audio signal (e.g., voice input **502**, voice input **602**, voice input **702**) for a voice user interface (e.g., VUI system **500**, VUI system **700**) of the wearable head device. Multi-microphone configurations can provide spatial information about a sound source in addition to audio information. For example, signal processing techniques can be used to determine a relative position of an audio source to wearable head device **100B** based on the amplitudes of the signals received at the multi-microphone configuration. If the same audio signal is received with a larger amplitude at microphone **150A** than at **150B**, it can be determined that the audio source is closer to microphone **150A** than to microphone **150B**. Asymmetric or symmetric microphone configurations can be used. In some embodiments, it can be advantageous to asymmetrically configure microphones **150A** and **150B** on a front face of wearable head device **100B**. For example, an asymmetric configuration of microphones **150A** and **150B** can provide spatial information pertaining to height (e.g., a distance from a first microphone to a voice source (e.g., the user's mouth, the user's throat) and a second distance from a second microphone to the voice source are different). This can be used to distinguish a user's speech from other human speech. For example, a ratio of amplitudes received at microphone **150A** and at microphone **150B** can be expected for a user's mouth to determine that an audio source is from the user. In some embodiments, a symmetrical configuration may be able to distinguish a user's speech from other human speech to the left or right of a user.

Although four microphones are shown in FIG. **1B**, it is contemplated that any suitable number of microphones can be used, and the microphone(s) can be arranged in any suitable (e.g., symmetrical or asymmetrical) configuration.

(12) FIG. **2** illustrates an example mobile handheld controller component **200** of an example wearable system. In some examples, handheld controller **200** may be in wired or wireless communication with wearable head device **100A** and/or **100B** and/or auxiliary unit **300** described below. In some examples, handheld controller **200** includes a handle portion **220** to be held by a user, and one or more buttons **240** disposed along a top surface **210**. In some examples, handheld controller **200** may be configured for use as an optical tracking target; for example, a sensor (e.g., a camera or other optical sensor) of wearable head device **100A** and/or **100B** can be configured to detect a position and/or orientation of handheld controller **200** which may, by extension, indicate a position and/or orientation of the hand of a user holding handheld controller **200**. In some examples, handheld controller **200** may include a processor, a memory, a storage unit, a display, or one or more input devices, such as ones described herein. In some examples, handheld controller **200** includes one or more sensors (e.g., any of the sensors or tracking components described herein with respect to wearable head device **100A** and/or **100B**). In some examples, sensors can detect a position or orientation of handheld controller **200** relative to wearable head device **100A** and/or **100B** or to another component of a wearable system. In some examples, sensors may be positioned in handle portion **220** of handheld controller **200**, and/or may be mechanically coupled to the handheld controller. Handheld controller **200**

can be configured to provide one or more output signals, corresponding, for example, to a pressed state of the buttons **240**; or a position, orientation, and/or motion of the handheld controller **200** (e.g., via an IMU). Such output signals may be used as input to a processor of wearable head device **100A** and/or **100B**, to auxiliary unit **300**, or to another component of a wearable system. In some examples, handheld controller **200** can include one or more microphones to detect sounds (e.g., a user's speech, environmental sounds), and in some cases provide a signal corresponding to the detected sound to a processor (e.g., a processor of wearable head device **100A** and/or **100B**).

(13) FIG. **3** illustrates an example auxiliary unit **300** of an example wearable system. In some examples, auxiliary unit **300** may be in wired or wireless communication with wearable head device **100A** and/or **100B** and/or handheld controller **200**. The auxiliary unit **300** can include a battery to primarily or supplementally provide energy to operate one or more components of a wearable system, such as wearable head device **100A** and/or **100B** and/or handheld controller **200** (including displays, sensors, acoustic structures, processors, microphones, and/or other components of wearable head device **100A** and/or **100B** or handheld controller **200**). In some examples, auxiliary unit **300** may include a processor, a memory, a storage unit, a display, one or more input devices, and/or one or more sensors, such as ones described herein. In some examples, auxiliary unit **300** includes a clip **310** for attaching the auxiliary unit to a user (e.g., attaching the auxiliary unit to a belt worn by the user). An advantage of using auxiliary unit **300** to house one or more components of a wearable system is that doing so may allow larger or heavier components to be carried on a user's waist, chest, or back—which are relatively well suited to support larger and heavier objects—rather than mounted to the user's head (e.g., if housed in wearable head device **100A** and/or **100B**) or carried by the user's hand (e.g., if housed in handheld controller **200**). This may be particularly advantageous for relatively heavier or bulkier components, such as batteries.

(14) FIG. **4A** shows an example functional block diagram that may correspond to an example wearable system **401A**; such system may include example wearable head device **100A** and/or **100B**, handheld controller **200**, and auxiliary unit **300** described herein. In some examples, the wearable system **401A** could be used for virtual reality, augmented reality, or mixed reality applications. As shown in FIG. **4**, wearable system **401A** can include example handheld controller **400B**, referred to here as a “totem” (and which may correspond to handheld controller **200**); the handheld controller **400B** can include a totem-to-headgear six degree of freedom (6DOF) totem subsystem **404A**. Wearable system **401A** can also include example headgear device **400A** (which may correspond to wearable head device **100A** and/or **100B**); the headgear device **400A** includes a totem-to-headgear 6DOF headgear subsystem **404B**. In the example, the 6DOF totem subsystem **404A** and the 6DOF headgear subsystem **404B** cooperate to determine six coordinates (e.g., offsets in three translation directions and rotation along three axes) of the handheld controller **400B** relative to the headgear device **400A**. The six degrees of freedom may be expressed relative to a coordinate system of the headgear device **400A**. The three translation offsets may be expressed as X, Y, and Z offsets in such a coordinate system, as a translation matrix, or as some other representation. The rotation degrees of freedom may be expressed as sequence of yaw, pitch and roll rotations; as vectors; as a rotation matrix; as a quaternion; or as some other representation. In some examples, one or more depth cameras **444** (and/or one or more non-depth cameras) included in the headgear device **400A**; and/or one or more optical targets (e.g., buttons **240** of handheld controller **200** as described, dedicated optical targets included in the handheld controller) can be used for 6DOF tracking. In some examples, the handheld controller **400B** can include a camera, as described; and the headgear device **400A** can include an optical target for optical tracking in conjunction with the camera. In some examples, the headgear device **400A** and the handheld controller **400B** each include a set of three orthogonally oriented solenoids which are used to wirelessly send and receive three distinguishable signals. By measuring the relative magnitude of the three distinguishable signals received in each of the coils used for receiving, the 6DOF of the handheld controller **400B** relative to the headgear device **400A** may be determined. In some examples, 6DOF totem subsystem **404A** can include an Inertial Measurement Unit (IMU) that is useful to provide improved accuracy and/or more timely information on rapid movements of the handheld controller **400B**.

(15) FIG. **4B** shows an example functional block diagram that may correspond to an example wearable system **401B** (which can correspond to example wearable system **401A**). In some embodiments, wearable system **401B** can include microphone array **407**, which can include one or more microphones arranged on headgear device **400A**. In some embodiments, microphone array **407** can include four microphones. Two microphones can be placed on a front face of headgear **400A**, and two microphones can be placed at a rear of head headgear **400A** (e.g., one at a back-left and one at a back-right), such as the configuration described with respect to FIG. **1B**. The microphone array **407** can include any suitable number of microphones, and can include a single microphone. In some embodiments, signals received by microphone array **407** can be transmitted to DSP **408**. DSP **408** can be configured to perform signal processing on the signals received from microphone array **407**. For example, DSP **408** can be configured to perform noise reduction, acoustic echo cancellation, and/or beamforming on signals received from microphone array **407**. DSP **408** can be configured to transmit signals to processor **416**. In some embodiments, the DSP **408** and/or processor **416** are configured to execute instructions stored in memory of the wearable system to implement aspects of a voice user interface system (e.g., VUI system **500**, VUI system **700**) configured to receive a voice input or an

audio signal from microphone array **407**. In some embodiments, the system **401B** can include multiple signal processing stages that may each be associated with one or more microphones. In some embodiments, the multiple signal processing stages are each associated with a microphone or a combination of two or more microphones used for beamforming. In some embodiments, the multiple signal processing stages are each associated with noise reduction or echo-cancellation algorithms used to pre-process a signal used for either voice onset detection, key phrase detection, or endpoint detection.

(16) In some examples involving augmented reality or mixed reality applications, it may be desirable to transform coordinates from a local coordinate space (e.g., a coordinate space fixed relative to headgear device **400A**) to an inertial coordinate space, or to an environmental coordinate space. For instance, such transformations may be necessary for a display of headgear device **400A** to present a virtual object at an expected position and orientation relative to the real environment (e.g., a virtual person sitting in a real chair, facing forward, regardless of the position and orientation of headgear device **400A**), rather than at a fixed position and orientation on the display (e.g., at the same position in the display of headgear device **400A**). This can maintain an illusion that the virtual object exists in the real environment (and does not, for example, appear positioned unnaturally in the real environment as the headgear device **400A** shifts and rotates). In some examples, a compensatory transformation between coordinate spaces can be determined by processing imagery from the depth cameras **444** (e.g., using a Simultaneous Localization and Mapping (SLAM) and/or visual odometry procedure) in order to determine the transformation of the headgear device **400A** relative to an inertial or environmental coordinate system. In the example shown in FIG. **4**, the depth cameras **444** can be coupled to a SLAM/visual odometry block **406** and can provide imagery to block **406**. The SLAM/visual odometry block **406** implementation can include a processor configured to process this imagery and determine a position and orientation of the user's head, which can then be used to identify a transformation between a head coordinate space and a real coordinate space. Similarly, in some examples, an additional source of information on the user's head pose and location is obtained from an IMU **409** of headgear device **400A**. Information from the IMU **409** can be integrated with information from the SLAM/visual odometry block **406** to provide improved accuracy and/or more timely information on rapid adjustments of the user's head pose and position.

(17) In some examples, the depth cameras **444** can supply 3D imagery to a hand gesture tracker **411**, which may be implemented in a processor of headgear device **400A**. The hand gesture tracker **411** can identify a user's hand gestures, for example by matching 3D imagery received from the depth cameras **444** to stored patterns representing hand gestures. Other suitable techniques of identifying a user's hand gestures will be apparent.

(18) In some examples, one or more processors **416** may be configured to receive data from headgear subsystem **404B**, the IMU **409**, the SLAM/visual odometry block **406**, depth cameras **444**, microphones **450**; and/or the hand gesture tracker **411**. The processor **416** can also send and receive control signals from the 6DOF totem system **404A**. The processor **416** may be coupled to the 6DOF totem system **404A** wirelessly, such as in examples where the handheld controller **400B** is untethered. Processor **416** may further communicate with additional components, such as an audio-visual content memory **418**, a Graphical Processing Unit (GPU) **420**, and/or a Digital Signal Processor (DSP) audio spatializer **422**. The DSP audio spatializer **422** may be coupled to a Head Related Transfer Function (HRTF) memory **425**. The GPU **420** can include a left channel output coupled to the left source of imagewise modulated light **424** and a right channel output coupled to the right source of imagewise modulated light **426**. GPU **420** can output stereoscopic image data to the sources of imagewise modulated light **424**, **426**. The DSP audio spatializer **422** can output audio to a left speaker **412** and/or a right speaker **414**. The DSP audio spatializer **422** can receive input from processor **419** indicating a direction vector from a user to a virtual sound source (which may be moved by the user, e.g., via the handheld controller **400B**). Based on the direction vector, the DSP audio spatializer **422** can determine a corresponding HRTF (e.g., by accessing a HRTF, or by interpolating multiple HRTFs). The DSP audio spatializer **422** can then apply the determined HRTF to an audio signal, such as an audio signal corresponding to a virtual sound generated by a virtual object. This can enhance the believability and realism of the virtual sound, by incorporating the relative position and orientation of the user relative to the virtual sound in the mixed reality environment—that is, by presenting a virtual sound that matches a user's expectations of what that virtual sound would sound like if it were a real sound in a real environment.

(19) In some examples, such as shown in FIG. **4**, one or more of processor **416**, GPU **420**, DSP audio spatializer **422**, HRTF memory **425**, and audio/visual content memory **418** may be included in an auxiliary unit **400C** (which may correspond to auxiliary unit **300**). The auxiliary unit **400C** may include a battery **427** to power its components and/or to supply power to headgear device **400A** and/or handheld controller **400B**. Including such components in an auxiliary unit, which can be mounted to a user's waist, can limit or reduce the size and weight of headgear device **400A**, which can in turn reduce fatigue of a user's head and neck.

(20) While FIGS. **4A** and **4B** present elements corresponding to various components of an example wearable systems **401A** and **401B**, various other suitable arrangements of these components will become apparent to those skilled in the art. For example, the headgear device **400A** illustrated in FIG. **4A** or FIG. **4B** may include a processor

and/or a battery (not shown). The included processor and/or battery may operate together with or operate in place of the processor and/or battery of the auxiliary unit **400C**. Generally, as another example, elements presented or functionalities described with respect to FIG. 4 as being associated with auxiliary unit **400C** could instead be associated with headgear device **400A** or handheld controller **400B**. Furthermore, some wearable systems may forgo entirely a handheld controller **400B** or auxiliary unit **400C**. Such changes and modifications are to be understood as being included within the scope of the disclosed examples.

(21) FIG. 5 illustrates an exemplary voice user interface (VUI) system **500** according to some embodiments of the disclosure. The VUI system **500** may be implemented on a system such as wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**, and a mixed reality system. However, it is understood that the described wearable systems are merely exemplary, and that the VUI system may be implemented in other systems (e.g., a voice assistant running on a device connected to a power outlet or a voice assistant in a vehicle). Instructions associated with elements of the VUI system may be stored in memories of a system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**, a mixed reality system) comprising the VUI system and executed by a processor (e.g., CPU, GPU, DSP, ASIC) of the system. Elements of the VUI system may also be implemented over different devices (e.g., some analyses may be performed over the cloud or over a neural network).

(22) In some embodiments, the VUI system **500** receives a voice input **502**, which may be an audio signal (e.g., using a microphone disclosed herein), and causes a system (e.g., mixed reality system) to perform an action (e.g., actions **516**, **518**, **520**, **530**, **532**) associated with the voice input. In some embodiments, the voice input may comprise a linguistic input or a non-linguistic input. As used herein, a linguistic input may be an input comprising a word of a language, and a non-linguistic input may be an input that does not substantially comprise a linguistic input (e.g., paralinguistic input, prosodic input, voice tones). In addition, the actions may be performed additionally based on a non-voice input (e.g., user movement, user facial expressions, user gestures) received by a sensor of the system as described herein.

(23) In some embodiments, the VUI may verify whether an input is a valid voice input before further processing the voice input (e.g., verify whether the input is a voice input of a user of the VUI, verify whether the input is directed to the VUI, verify whether the input is associated with an action). For example, a plurality of microphones (e.g., microphones **150**) may receive the audio input and determine a location of a source of the audio input based on levels and directions of the received signals at each of the microphones (e.g., as described with respect to FIGS. 1-4). In accordance with a determination that the source of the audio is associated with a user of the VUI, the voice input can be further processed, as disclosed herein.

(24) In some embodiments, a first processor (e.g., a CPU, GPU, DSP, ASIC) may determine whether the audio comprises a valid voice input (e.g., verify whether the input is a voice input of a user of the VUI, verify whether the input is directed to the VUI, verify whether the input is associated with an action) and a non-linguistic input, using the systems and methods described herein, before a second processor (e.g., CPU, GPU, DSP, ASIC) further processes the voice input with features of the disclosed VUI.

(25) Such capabilities may be advantageous in mobile applications of speech processing, such as virtual reality or augmented reality applications on a wearable device. In such wearable applications, the user may speak without directing input speech to the wearable system. The user may also be in locations where significant amounts of background speech exists. Further, the wearable system may be battery-operated and have a limited operation time. Sensors of wearable systems (such as those described herein with respect to FIGS. 1-4) may be advantageously utilized to solve the problems of speech processing inaccuracy (e.g., reduce false triggering of the VUI) and power consumption by verifying the validity of an audio input before further processing the input.

(26) In some embodiments, the voice input **502** may comprise more than one linguistic or non-linguistic input and can be segmented based on the voice input's temporal or spectral characteristics (e.g., pauses, change in temporal patterns, change in spectral patterns). In some examples, each of the segmented portions in the voice input may be processed independently in the VUI to determine a corresponding action. In some examples, each of the segmented portions may be processed independently and the kind of input for each of the portions is determined. The determined inputs may collected determine a corresponding action.

(27) In some embodiments, the VUI system **500** includes a voice input discriminator **504**, a convolutional neural network (CNN) **508**, non-linguistic feature computation engine **510**, non-linguistic feature classifier **512**, decision engine **514**, natural language understanding (NLU) engine **526**, and automatic speech recognition (ASR) engine **528**. Although the exemplary VUI system **500** is described with the illustrated elements, it is understood that other configurations (e.g., less elements, additional components) of VUI system **500** exist without departing from the scope of the disclosure.

(28) The voice input discriminator **504** may receive the voice input **502** (e.g., an audio signal associated with a voice input). It may determine whether the audio signal comprises a linguistic input **524** and/or a non-linguistic input **506**. Features of the voice input discriminator **504** and determination of whether a voice input or an audio signal

comprises a linguistic input **524** and/or a non-linguistic input **506** are described in more detail with respect to FIG. 6.

(29) In some embodiments, in accordance with a determination that the voice input **502** comprises a non-linguistic input **506** (e.g., a paralinguistic input, prosodic input), the VUI system **500** receives information associated with the voice input **502** or audio signal comprising the non-linguistic input **506** from the CNN **508**.

(30) In some embodiments, in accordance with a determination that the voice input **502** comprises a non-linguistic input **506** (e.g., a paralinguistic input, prosodic input), the non-linguistic feature computation engine **510** extracts non-linguistic features from the voice input **502** or audio signal comprising the non-linguistic input **506**. For example, the feature computation engine **510** may identify or extract time-domain and frequency-domain features of a non-linguistic input **506**. One or more these features may be used to uniquely identify a kind of non-linguistic input **506**.

(31) After the non-linguistic features had been extracted from the voice input **502** or audio signal, the extract features may be classified with the non-linguistic feature classifier **512**. From the one or a combination of these features, the non-linguistic feature classifier **512** may uniquely identify and classify the kind of non-linguistic input.

(32) In some embodiments, the decision engine **514** may receive the information from the CNN **508** and/or classification of features associated with the voice input **502** or audio signal comprising the non-linguistic input **506** from the non-linguistic feature classifier **512**. Based on the information from the CNN **508** and/or the feature classifications, an action associated with the voice input **502** may be determined. In response, the system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**, a mixed reality system) comprising the VUI system **500** may be caused to perform the determined action.

(33) For example, the determined non-linguistic input may be matched with a pre-configured action. In some embodiments, the pre-configured action may be mapped to a non-linguistic input by training the system. In some embodiments, the pre-configured action may be mapped by a designer of the VUI system to improve user accessibility to operations of the device implementing the VUI system. These actions are described in more detail herein.

(34) In some embodiments, the action comprises one of texting **516**, performing an intent **518**, or inserting an emoji **520**. For example, a vocal click may instruct the VUI to cause a device to take a picture (e.g., performing an intent **518**). As another example, two vocal clicks may instruct the VUI to cause a device to take a video (e.g., performing an intent **518**). As yet another example, while browsing a social media application, a user's "kiss" sound may cause a device to "like" a post and insert a smile or heart emoji (e.g., performing an intent **518** and inserting an emoji **520**). As yet another example, a laugh or a chuckle may cause a device to insert a smile or laughing emoji into a message (e.g., inserting an emoji **520**).

(35) In some embodiments, the action texting **516**, performing an intent **518**, or inserting an emoji **520** may be alternatively or additionally based on non-voice inputs. In some embodiments, a sensor of a device (e.g., a camera, any sensor disclosed herein) senses a facial expression of a user of the device to cause the device to perform the action. For example, a camera of the device may sense the user smiling and in accordance with a determination that the user is smiling, the device is caused to perform an associated action (e.g., insert a smiling emoji, take a selfie). As another example, a camera of the device may sense that the user is grimacing and in accordance with a determination that the user is grimacing, the device is caused to perform an associated action (e.g., insert a disgusted emoji, ask the user if medical assistance is needed). In some embodiments, a sensor of a device (e.g., a camera, an IMU, an accelerometer, any sensor disclosed herein) senses a gesture or movement of a user of the device to cause the device to perform the action. For example, a sensor of the device may sense that the user is waving and in accordance with a determination that the user is waving, the device is caused to perform an associated action (e.g., insert a waving emoji, send a greeting message). As another example, a sensor of the device may sense that the user is nodding his or her head and in accordance with a determination that the user is nodding, the device is caused to perform an associated action (e.g., replying yes in a message, inputting a confirmation).

(36) As an exemplary advantage, associating these non-linguistic inputs with an action results in a VUI that is able to respond to time-critical commands much closer to real-time than is possible with natural-language processing (NLP) systems. For example, the disclosed VUI may be advantageously used in healthcare applications, where execution of voice commands in a prompt manner may be critical (e.g., during a surgery, when a surgeon's hands are occupied and a hands-free and time-critical are needed).

(37) In some embodiments, a VUI may be trained to recognize a user's unique non-linguistic utterances (e.g., vocal clicks, pops, kisses). These utterances may then be used as combined WuWs and actions (e.g., actions **516**, **518**, **520**, **716**, **718**, **720**). In some embodiments, a user may map a recognized utterance to an action (e.g., mapping one click to taking a picture, mapping two clicks to recording a video). In some embodiments, more than one user may have different mappings between utterances and corresponding actions. For example, a first user may map one click to taking a picture, while a second user may map one click to recording a video. As another example, a first user may map a kissing sound to sending a heart emoji, while a second user may map a kissing sound to sending "I love you" to the user's mother.

(38) In some embodiments, a position of the device may be determined and the actions associated with the non-linguistic input are associated with the position of the system. For example, actions associated with the non-linguistic inputs may be culture specific, including based on the location (e.g., country) of the device.

(39) Although actions associated with the non-linguistic input **506** are described using actions **516**, **518**, and **520**, it is understood that these actions are merely exemplary and other actions associated with the non-linguistic input **506** exist without departing from the scope of the disclosure.

(40) In some embodiments, in accordance with a determination that the voice input **502** comprises a linguistic input **524**, the NLU engine **526** receives the voice input **502** or audio signal comprising the linguistic input **524**. The NLU engine **526** may determine an action associated with the voice input **502** or audio signal comprising the linguistic input **524**. The action may be intent **530**. In some embodiments, the NLU engine **526** is configured to receive the linguistic input **524** after a WuW is inputted.

(41) For example, a user may provide a voice input by saying “take a picture.” The voice user voice input discriminator **504** may determine that the voice input is a linguistic input. The NLU engine **526** receives the voice input and determine that the voice input comprises the words “take a picture.” Based on the words determined by the NLU engine, the VUI system determines that the words are associated with taking a picture with a camera. Based on this determination, the system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**, a mixed reality system) comprising the VUI would be caused to take a picture with the camera of the system.

(42) As illustrated in the examples, by processing non-linguistic information (e.g., tone, prosody, speech cadence, emotion) in a voice input, the disclosed voice user interface system may advantageously allow for quicker response and action on a device for time-critical actions.

(43) In some embodiments, in accordance with a determination that the voice input **502** comprises a linguistic input **524**, the ASR engine **528** receives the voice input **502** or audio signal comprising the linguistic input **524**. The ASR engine **528** may determine an action associated with the voice input **502** or audio signal comprising the linguistic input **524**. The action may be text **532**. In some embodiments, the ASR engine **528** is configured to receive the linguistic input **524** after a WuW and/or a messaging command are inputted.

(44) For example, a user may provide a voice input by saying “hi, I am on my way.” The voice user voice input discriminator **504** may determine that the voice input is a linguistic input. The ASR engine **528** receives the voice input and determine that the voice input comprises the words “hi, I am on my way.” Based on the words determined by the ASR engine, the VUI system determines that the words are associated with a message intended to be transmitted to a recipient. Based on this determination, the system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**, a mixed reality system) comprising the VUI would be caused to send the message “hi, I am on my way” to a desired recipient.

(45) Although actions associated with the non-linguistic input **524** are described using actions **530** and **532**, it is understood that these actions are merely exemplary and other actions associated with the non-linguistic input **524** exist without departing from the scope of the disclosure.

(46) FIG. **6** illustrates an exemplary voice input discriminator **600** according to some embodiments of the disclosure. In some embodiments, the voice input discriminator **600** is voice input discriminator **504** or voice input discriminator **704**. In some embodiments, the voice input discriminator **600** includes frequency-domain feature identifier **604**, time-domain feature identifier **606**, and voice input classifier **608**. The frequency-domain feature identifier **604** may quantify frequency-domain characteristics of the voice input **602** (e.g., voice input **502**, voice input **702**) (e.g., perform a spectral analysis on the voice input, determine fundamental frequency of the voice input). The time-domain feature identifier **606** may quantify time-domain characteristics of the voice input **602** (e.g., voice input **502**, voice input **702**) (e.g., measure amplitude of the voice input over time).

(47) The identified frequency and time-domain features of the voice input **602** may be send to the voice input classifier **608**. The voice input classifier **608** may communicate with a feature database **610**. The feature database **610** may include information (frequency-domain and time-domain features) associated with linguistic and non-linguistic (e.g., paralinguistic) inputs and the voice input classifier **608** may receive the information and, based on the received information, determine whether a voice input (e.g., voice input **502**, voice input **702**) comprises a linguistic input or a non-linguistic input.

(48) A non-linguistic input may include frequency-domain characteristics such as distinct frequency component and unchanging spectral content, which may be identified using the frequency-domain feature identifier **604**. For example, some non-linguistic input may include a high frequency component not typically found in linguistic inputs. As another example, non-linguistic inputs such as a grunt, a shrug, or an ugh may include low frequency component not typically found in linguistic inputs. Additionally, the frequency-domain feature identifier **604** may identify linguistic spectral characteristics such as formants in a voice input. A non-linguistic input (e.g., a click, a kissing sound) may include transient or less dynamic (e.g., less change over time, compared to a linguistic input) time-domain characteristics, which may be identified using the time-domain feature identifier **606**. The time-domain

feature identifier **606** may also identify linguistic temporal characteristics such as rhythm and rate in a voice input. (49) Based on the identified time-domain and frequency-domain features using the frequency-domain feature identifier **604** and time-domain feature identifier **606**, the feature database **610** may transmit information related to the identified features to the voice input classifier **608**. The voice input classifier **608** may determine whether the voice input **602** comprises a linguistic input and/or a non-linguistic input and transmit based on the information received from the feature database **610**.

(50) In some embodiments, the feature database **610** may be part of a neural network. In some embodiments, the feature database **610** may be a database external to a device implementing the VUI system (e.g., cloud). In some embodiments, the feature database **610** may be stored in a device (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**) comprising the VUI.

(51) FIG. 7 illustrates an exemplary VUI system **700** according to some embodiments of the disclosure. The VUI system **700** may be implemented on a system such as wearable head device **100A**, wearable head device **100B**, wearable system **401A**, and wearable system **401B**. However, it is understood that the described wearable systems are merely exemplary, and that the VUI system may be implemented in other systems. Instructions associated with elements of the VUI system may be stored in memories of a system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**) comprising the VUI system and executed by a processor (e.g., CPU, GPU, DSP, ASIC) of the system. Elements of the VUI system may also be implemented over different devices (e.g., some analyses may be performed over the cloud).

(52) In some embodiments, the VUI system **700** receives a voice input **702** (e.g., using a microphone disclosed herein) and causes a system (e.g., mixed reality system) to perform an action (e.g., actions **716**, **718**, **720**, **730**, **732**) associated with the voice input. In some embodiments, the voice input may comprise a linguistic input or a non-linguistic input.

(53) In some embodiments, the VUI system **700** includes a voice input discriminator **704**, a convolutional neural network (CNN) **708**, non-linguistic feature computation engine **710**, non-linguistic feature classifier **712**, decision engine **714**, natural language understanding (NLU) engine **726**, and automatic speech recognition (ASR) engine **728**. Although the exemplary VUI system **700** is described with the illustrated elements, it is understood that other configurations (e.g., less elements, additional components) of VUI system **500** exist without departing from the scope of the disclosure.

(54) In some embodiments, some of the elements in VUI system **700** may be substantially similar to elements in VUI system **500**. For example, the voice input discriminator **704** may be substantially similar to the voice input discriminator **504**, the CNN **708** may be substantially similar to CNN **508**, the non-linguistic feature computation engine **710** may be substantially similar to non-linguistic feature computation engine **510**, non-linguistic feature classifier **712** may be substantially similar to non-linguistic feature classifier **512**, decision engine **714** may be substantially similar to decision engine **514**, NLU engine **726** may be substantially similar to NLU engine **526**, and ASR engine **728** may be substantially similar to ASR engine **528**, the actions **716**, **718**, and **720** may be substantially similar to actions **516**, **518**, and **520**.

(55) In some embodiments, as illustrated, CNN **708** and feature engine computation **710** form non-linguistic analysis unit **734**. The non-linguistic analysis unit **734** may provide information to NLU engine **726** and ASR engine **728**. The provided information may yield more accurate results associated with intent action **730** or text action **732** associated with a linguistic input of the voice input **702**. For example, the non-linguistic analysis unit **734** may provide more information about a voice input comprising linguistic input beyond merely the words of the voice input, such as conveying emotion and intent.

(56) In some embodiments, the voice input **702** may comprise linguistic and non-linguistic input that may affect an associated action (e.g., actions **730**, **732**). For example, a user may provide a voice input by saying “how are you?”. The voice user voice input discriminator **704** may determine that the voice input comprises a linguistic input and a non-linguistic input. The ASR engine **728** receives the voice input and determine that the voice input comprises the words “how are you.” The non-linguistic analysis unit **734** may determine that the voice input comprises a non-linguistic input of a raised tone near the end of the voice input. Based on the words determined by the ASR engine and the determination of a raised tone in the voice input, the VUI system determines that the words are associated with a message intended to be transmitted to a recipient in the form of a question. Based on these determinations, the system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**, a mixed reality system) comprising the VUI would be caused to send the message “how are you?” to a desired recipient.

(57) The non-linguistic analysis unit **734** may determine that the voice input comprises a non-linguistic input comprising inflection and prosody. These non-linguistic input may emphasize more important portions of the voice input, and the associated action may be performed based on these non-linguistic input (e.g., emphasis made be added to the more important portions of a message, an action may be performed with a higher priority).

(58) For example, a user may provide a voice input by saying “WHEN is Mom's birthday?”. The voice user voice

input discriminator **704** may determine that the voice input comprises a linguistic input and a non-linguistic input. The ASR engine **728** receives the voice input and determine that the voice input comprises the words “when is Mom's birthday.” The non-linguistic analysis unit **734** may determine that the voice input comprises a non-linguistic input comprising an emphasis on the word “when.” Based on the words determined by the ASR engine and the determination of a raised tone and the emphasis in the voice input, the VUI system determines that the words are associated with a message intended to be transmitted to a recipient in the form of a question and the word “when” is emphasized. Based on these determinations, the system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**, a mixed reality system) comprising the VUI would be caused to send the message “WHEN is Mom's birthday?” to a desired recipient. Alternatively, based on these determinations, in the message, the word “when” may be formatted (e.g., boldface, italics, underline, underline, strikethrough, diacritical marks, punctuation formatting, font is enlarged, a different font is used) to indicate emphasis on the word.

(59) As another example, a user may provide a voice input by saying “Ha. Ha.” in a sarcastic manner. The voice user voice input discriminator **704** may determine that the voice input comprises a linguistic input and a non-linguistic input. The ASR engine **728** receives the voice input and determine that the voice input comprises the words “Haha.” The non-linguistic analysis unit **734** may determine that the voice input comprises a non-linguistic input comprising a sarcastic tone (e.g., by comparing time-domain and frequency-domain features of the input with features of typical usage of the input words). Based on the words determined by the ASR engine and the determination of a sarcastic tone in the voice input, the VUI system determines that the words are associated with a message intended to be transmitted to a recipient to convey sarcasm. Based on these determinations, the system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**, a mixed reality system) comprising the VUI would be caused to send the message “Ha. Ha.” to a desired recipient. Alternatively, the system may be caused to send the message along with a smiley or winking emoji or an indicator (e.g., “Is,” “/sarcasm” to convey sarcasm in the message.

(60) As yet another example, a user may provide a voice input by saying “TAKE A PICTURE” in a louder voice. The voice user voice input discriminator **704** may determine that the voice input comprises a linguistic input and a non-linguistic input. The NLU engine **726** receives the voice input and determine that the voice input comprises the words “take a picture,” causing a device comprising the VUI to take a picture. The non-linguistic analysis unit **734** may determine that the voice input comprises a non-linguistic input comprising an emphasis on the entire input. Based on the words determined by the NLU engine and the determination of the emphasis in the voice input, the VUI system determines that it should cause a picture to be taken, but the picture taking operation should be modified. For example, the emphasis of the input may convey urgency for taking the picture. Based on these determinations, the system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**, a mixed reality system) comprising the VUI would be caused the picture to be taken in a modified manner (e.g., the picture may be taken more quickly, a confirmation prompt may be bypassed to take the picture in a quicker manner, a default setting may be used to take the picture without prompting the user, system resources may be freed up to prioritize the picture taking task).

(61) As illustrated in the examples, by processing non-linguistic information (e.g., tone, prosody, speech cadence, emotion) in a voice input, the disclosed voice user interface system may advantageously convey subtleties of a user's underlying emotion and/or intent by supplementing a user's text with appropriate emojis, formatting a user's text to assist in conveyance of the user's emotion and tone, and/or altering the user's supplied text.

(62) Although actions associated with the non-linguistic input **706** are described using actions **716**, **718**, and **720**, it is understood that these actions are merely exemplary and other actions associated with the non-linguistic input **706** exist without departing from the scope of the disclosure. Although actions associated with the non-linguistic input **724** are described using actions **730** and **732**, it is understood that these actions are merely exemplary and other actions associated with the non-linguistic input **724** exist without departing from the scope of the disclosure.

(63) FIG. **8** illustrates an exemplary method **800** of operating a VUI system according to some embodiments of the disclosure. Although the method **800** is illustrated as including the described steps, it is understood that different order of step, additional step, or less step may be included without departing from the scope of the disclosure. In some embodiments, the method **800** is performed with at least one of VUI system **500** and VUI system **700**. For the sake of brevity, some elements and advantages associated with these systems are not repeated here.

(64) In some embodiments, the method **800** includes receiving, using a microphone of a system, an audio signal (step **802**). For example, as described with respect to FIG. **5**, the VUI system **500** receives a voice input **502** using microphone of the system. As another example, as described with respect to FIG. **7**, the VUI system **700** receives a voice input **702** using microphone of the system.

(65) In some embodiments, the method **800** includes determining whether the audio signal comprises a non-linguistic input (step **804**). For example, as described with respect to FIG. **5**, the voice input discriminator **504** receives the voice input **502** and determine whether the voice input comprises a linguistic input **524** and/or a non-

linguistic input **506**. As another example, as described with respect to FIG. 7, the voice input discriminator **704** receives the voice input **702** and determine whether the voice input comprises a linguistic input **724** and/or a non-linguistic input **706**.

(66) In some embodiments, the method **800** includes in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input (step **806**). For example, as described with respect to FIG. 5, the VUI system **500** causes an action (e.g., actions **516**, **518**, **520**) associated with a non-linguistic input **506** to be performed. As another example, as described with respect to FIG. 7, the VUI system **700** causes an action (e.g., actions **716**, **718**, **720**) associated with a non-linguistic input **706** to be performed.

(67) In some embodiments, the action comprises one of texting, performing an intent, and inserting an emoji. For example, as described with respect to FIG. 5, the action comprises one of texting **516**, performing an intent **518**, and inserting an emoji **520**. As another example, as described with respect to FIG. 7, the action comprises one of texting **716**, performing intent **718**, and inserting an emoji **720**.

(68) In some embodiments, the system is a mixed reality system in a mixed reality environment, and the action is further associated with the mixed reality environment. For example, the mixed reality system (e.g., wearable head device **100A**, wearable head device **100B**, wearable system **401A**, wearable system **401B**) includes the VUI system **500** and/or the VUI system **700** and is in a mixed reality environment. The action that the VUI system **500** and/or the VUI system **700** causes to perform is an action associated with the mixed reality environment.

(69) In some embodiments, the non-linguistic input is a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input. For example, the

(70) In some embodiments, the non-linguistic input is a prosodic input, and in accordance with a determination that the audio signal comprises the prosodic input, the action comprises a second action associated with the prosodic input. For example, as described with respect to FIG. 5, the voice input discriminator **504** determines that the voice input includes a prosodic input, and the VUI system **500** causes an action associated with the prosodic input, as described herein, to be performed. As another example, as described with respect to FIG. 7, the voice input discriminator **704** determines that the voice input includes a prosodic input, and the VUI system **700** causes an action associated with the prosodic input, as described herein, to be performed.

(71) In some embodiments, the second action is a modification of an action associated with a linguistic input. For example, as described with respect to FIG. 7, the voice input discriminator **704** determines that the voice input includes a prosodic input, and the VUI system **700** causes an action associated with a linguistic input to be modified, as described herein.

(72) In some embodiments, the prosodic input is indicative of an emotion associated with a user of the system, the user of the system associated with the audio signal, and the action is further associated with the emotion. For example, as described with respect to FIGS. 5 and 7, a voice input is determined to include a prosodic input, and the prosodic input is indicative of an emotion of a user who provided an input to the VUI system **500** or VUI system **700**. The action associated with prosodic input is further associated with the emotion of the user, as described herein.

(73) In some embodiments, the method **800** includes determining whether the audio signal comprises a linguistic input; and in accordance with a determination that the audio signal comprises the linguistic input, performing a third action associated with the linguistic input. For example, as described with respect to FIG. 5, the voice input discriminator **504** determines the voice input comprises a linguistic input **524**, and the VUI system **500** causes an action (e.g., actions **530**, **532**) associated with a linguistic input **524** to be performed. As another example, as described with respect to FIG. 7, the voice input discriminator **704** determines the voice input comprises a linguistic input **724**, and the VUI system **700** causes an action (e.g., actions **730**, **732**) associated with a linguistic input **724** to be performed.

(74) In some embodiments, the action comprises a modification of the third action based on the non-linguistic input. For example, as described with respect to FIG. 7, the action (e.g., actions **730**, **732**) associated with a linguistic input **724** is modified, as described herein, based on the non-linguistic input **706**.

(75) In some embodiments, the method **800** includes in accordance with a determination that the audio signal comprises the non-linguistic input, receiving information associated with the audio signal from a convolutional neural network, wherein the action is performed based on the information. For example, as described with respect to FIG. 5, the voice input discriminator **504** determines that the voice input includes a non-linguistic input, and the convolutional neural network **508** receives information associated with the voice input for processing, as described herein. For example, as described with respect to FIG. 7, the voice input discriminator **704** determines that the voice input includes a non-linguistic input, and the convolutional neural network **708** receives information associated with the voice input for processing, as described herein.

(76) In some embodiments, the method **800** includes classifying a feature of the non-linguistic input. The action is performed based on the classified feature. For example, as described with respect to FIG. 5, non-linguistic feature

classifier **512** identifies and classifies a kind of non-linguistic input corresponding to the non-linguistic input, and the VUI system **500** causes an action to be performed based on the kind of non-linguistic input. As another example, as described with respect to FIG. 7, non-linguistic feature classifier **712** identifies and classifies a kind of non-linguistic input corresponding to the non-linguistic input, and the VUI system **700** causes an action to be performed based on the kind of non-linguistic input.

(77) In some embodiments, the method **800** includes receiving, from a sensor of the system different from the microphone, information associated with an environment of the system, wherein the action is further associated with the information received from the sensor. For example, as described with respect to FIG. 5 or FIG. 7, a sensor of a device including the VUI system **500** or the VUI system **700** receives an input including information associated with an environment of the system, and the VUI system **500** or the VUI system **700** causes the device to perform an action associated with the information associated with the environment of the system.

(78) In some embodiments, the method **800** includes determining a position of the system, wherein the action is further associated with the position of the system. For example, as described with respect to FIGS. 5 and 7, actions associated with the non-linguistic inputs may be culture specific, based on the location (e.g., country) of the device.

(79) In some embodiments, in accordance with a determination that the system is associated with a first user, the action comprises a first action associated with the first user; and in accordance with a determination that the system is associated with a second user, different from the first user, the action comprises a second action associated with the second user, different from the first action. For example, as described with respect to FIGS. 5 and 7, a first user may map one click to taking a picture, while a second user may map one click to recording a video. As another example, a first user may map a kissing sound to sending a heart emoji, while a second user may map a kissing sound to sending “I love you” to the user's mother.

(80) In some embodiments, the audio signal comprises a frequency-domain feature and a time-domain feature, and the determination of whether the audio comprises the non-linguistic input is based on the frequency-domain feature and the time-domain feature. For example, as described with respect to FIGS. 5 and 6, the voice input discriminator **502** or the voice input discriminator **602** identifies frequency-domain and time-domain features in the voice input, and the voice input discriminator **502** or the voice input discriminator **602** determines whether the voice input includes a non-linguistic input based on the identification of the frequency-domain and time-domain features.

(81) In some embodiments, the method **800** includes receiving information from a feature database, wherein the determination of whether the audio comprises the non-linguistic input is further based on the information. For example, as described with respect to FIG. 6, the VUI system receives information related to identified features from a feature database **610**, and the voice input discriminator **600** determines whether the voice input includes a non-linguistic input based on the information from the feature database **610**.

(82) In some embodiments, determining whether the audio signal comprises the non-linguistic input comprises using a first processor, and the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, waking up a second processor to perform the action. For example, a first processor (e.g., a CPU, GPU, DSP, ASIC) may determine whether the audio comprises a voice input (e.g., verify whether the input is a voice input of a user of the VUI, verify whether the input is directed to the VUI, verify whether the input is associated with an action) and a non-linguistic input, using the systems and methods described herein, before a second processor (e.g., CPU, GPU, DSP, ASIC) further processes the voice input with features of the disclosed VUI.

(83) According to some embodiments, a method comprises: receiving, using a microphone of a system, an audio signal; determining whether the audio signal comprises a non-linguistic input; and in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input.

(84) According to some embodiments, the non-linguistic input is a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input.

(85) According to some embodiments, the non-linguistic input is a prosodic input, and in accordance with a determination that the audio signal comprises the prosodic input, the action comprises a second action associated with the prosodic input.

(86) According to some embodiments, the second action is a modification of an action associated with a linguistic input.

(87) According to some embodiments, the prosodic input is indicative of an emotion associated with a user of the system, the user of the system associated with the audio signal, and the action is further associated with the emotion.

(88) According to some embodiments, the method further comprises: determining whether the audio signal comprises a linguistic input; and in accordance with a determination that the audio signal comprises the linguistic input, performing a third action associated with the linguistic input.

(89) According to some embodiments, the action comprises a modification of the third action based on the non-

linguistic input.

(90) According to some embodiments, the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, receiving information associated with the audio signal from a convolutional neural network, wherein the action is performed based on the information.

(91) According to some embodiments, the method further comprises classifying a feature of the non-linguistic input, wherein the action is performed based on the classified feature.

(92) According to some embodiments, the method further comprises associating the action with the non-linguistic input.

(93) According to some embodiments, the action comprises one of texting, performing an intent, and inserting an emoji.

(94) According to some embodiments, the method further comprises receiving, from a sensor of the system different from the microphone, information associated with an environment of the system, wherein the action is further associated with the information received from the sensor.

(95) According to some embodiments, the system is a mixed reality system in a mixed reality environment, and the action is further associated with the mixed reality environment.

(96) According to some embodiments, the method further comprises determining a position of the system, wherein the action is further associated with the position of the system.

(97) According to some embodiments, in accordance with a determination that the system is associated with a first user, the action comprises a first action associated with the first user; and in accordance with a determination that the system is associated with a second user, different from the first user, the action comprises a second action associated with the second user, different from the first action.

(98) According to some embodiments, the audio signal comprises a frequency-domain feature and a time-domain feature, and the determination of whether the audio comprises the non-linguistic input is based on the frequency-domain feature and the time-domain feature.

(99) According to some embodiments, the method further comprises receiving information from a feature database, wherein the determination of whether the audio comprises the non-linguistic input is further based on the information.

(100) According to some embodiments, determining whether the audio signal comprises the non-linguistic input comprises using a first processor, and the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, waking up a second processor to perform the action.

(101) According to some embodiments, a system comprises: a microphone; and one or more processors configured to execute a method comprising: receiving, using the microphone, an audio signal; determining whether the audio signal comprises a non-linguistic input; and in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input.

(102) According to some embodiments, the non-linguistic input is a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input.

(103) According to some embodiments, the non-linguistic input is a prosodic input, and in accordance with a determination that the audio signal comprises the prosodic input, the action comprises a second action associated with the prosodic input.

(104) According to some embodiments, the second action is a modification of an action associated with a linguistic input.

(105) According to some embodiments, the prosodic input is indicative of an emotion associated with a user of the system, the user of the system associated with the audio signal, and the action is further associated with the emotion.

(106) According to some embodiments, the method further comprises: determining whether the audio signal comprises a linguistic input; and in accordance with a determination that the audio signal comprises the linguistic input, performing a third action associated with the linguistic input.

(107) According to some embodiments, the action comprises a modification of the third action based on the non-linguistic input.

(108) According to some embodiments, the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, receiving information associated with the audio signal from a convolutional neural network, wherein the action is performed based on the information.

(109) According to some embodiments, the method further comprises classifying a feature of the non-linguistic input, wherein the action is performed based on the classified feature.

(110) According to some embodiments, the method further comprises associating the action with the non-linguistic input.

(111) According to some embodiments, the action comprises one of texting, performing an intent, and inserting an emoji.

(112) According to some embodiments, the system further comprises a sensor different from the microphone, wherein the method further comprises receiving, from the sensor, information associated with an environment of the system, wherein the action is further associated with the information received from the sensor.

(113) According to some embodiments, the system is a mixed reality system in a mixed reality environment, and the action is further associated with the mixed reality environment.

(114) According to some embodiments, the method further comprises determining a position of the system, wherein the action is further associated with the position of the system.

(115) According to some embodiments, in accordance with a determination that the system is associated with a first user, the action comprises a first action associated with the first user; and in accordance with a determination that the system is associated with a second user, different from the first user, the action comprises a second action associated with the second user, different from the first action.

(116) According to some embodiments, the audio signal comprises a frequency-domain feature and a time-domain feature, and the determination of whether the audio comprises the non-linguistic input is based on the frequency-domain feature and the time-domain feature.

(117) According to some embodiments, the method further comprises receiving information from a feature database, wherein the determination of whether the audio comprises the non-linguistic input is further based on the information.

(118) According to some embodiments, determining whether the audio signal comprises the non-linguistic input comprises using a first processor, and the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, waking up a second processor to perform the action.

(119) According to some embodiments, a non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to execute a method comprising: receiving, using a microphone of a system, an audio signal; determining whether the audio signal comprises a non-linguistic input; and in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input.

(120) According to some embodiments, the non-linguistic input is a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input.

(121) According to some embodiments, the non-linguistic input is a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input.

(122) According to some embodiments, the non-linguistic input is a prosodic input, and in accordance with a determination that the audio signal comprises the prosodic input, the action comprises a second action associated with the prosodic input.

(123) According to some embodiments, the second action is a modification of an action associated with a linguistic input.

(124) According to some embodiments, the prosodic input is indicative of an emotion associated with a user of the system, the user of the system associated with the audio signal, and the action is further associated with the emotion.

(125) According to some embodiments, the method further comprises: determining whether the audio signal comprises a linguistic input; and in accordance with a determination that the audio signal comprises the linguistic input, performing a third action associated with the linguistic input.

(126) According to some embodiments, the action comprises a modification of the third action based on the non-linguistic input.

(127) According to some embodiments, the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, receiving information associated with the audio signal from a convolutional neural network, wherein the action is performed based on the information.

(128) According to some embodiments, the method further comprises classifying a feature of the non-linguistic input, wherein the action is performed based on the classified feature.

(129) According to some embodiments, the method further comprises associating the action with the non-linguistic input.

(130) According to some embodiments, the action comprises one of texting, performing an intent, and inserting an emoji.

(131) According to some embodiments, the method further comprises receiving, from a sensor of the system different from the microphone, information associated with an environment of the system, wherein the action is further associated with the information received from the sensor.

(132) According to some embodiments, the system is a mixed reality system in a mixed reality environment, and the action is further associated with the mixed reality environment.

(133) According to some embodiments, the method further comprises determining a position of the system, wherein

the action is further associated with the position of the system.

(134) According to some embodiments, in accordance with a determination that the system is associated with a first user, the action comprises a first action associated with the first user; and in accordance with a determination that the system is associated with a second user, different from the first user, the action comprises a second action associated with the second user, different from the first action.

(135) According to some embodiments, the audio signal comprises a frequency-domain feature and a time-domain feature, and the determination of whether the audio comprises the non-linguistic input is based on the frequency-domain feature and the time-domain feature.

(136) According to some embodiments, the method further comprises receiving information from a feature database, wherein the determination of whether the audio comprises the non-linguistic input is further based on the information.

(137) According to some embodiments, determining whether the audio signal comprises the non-linguistic input comprises using a first processor, and the method further comprises in accordance with a determination that the audio signal comprises the non-linguistic input, waking up a second processor to perform the action.

(138) Although the disclosed examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. For example, elements of one or more implementations may be combined, deleted, modified, or supplemented to form further implementations. Such changes and modifications are to be understood as being included within the scope of the disclosed examples as defined by the appended claims.

Claims

1. A system comprising: a plurality of microphones of a wearable head device; and one or more processors configured to perform a method comprising: receiving, using the plurality of microphones, an audio signal; determining whether the audio signal is associated with a user of the wearable head device, wherein the determining whether the audio signal is associated with the user comprises: determining respective distances between a source of the audio signal and each of the plurality of microphones, and determining, based on the respective distances, a location of the source of the audio signal; in accordance with a determination that the audio signal is associated with the user of the wearable head device: determining whether the audio signal comprises a non-linguistic input; and in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input; and in accordance with a determination that the audio signal is not associated with the user of the wearable head device, forgoing determining whether the audio signal comprises the non-linguistic input.
2. The system of claim 1, wherein: the non-linguistic input comprises a paralinguistic input, and in accordance with a determination that the audio signal comprises the paralinguistic input, the action comprises a first action associated with the paralinguistic input.
3. The system of claim 1, wherein: the non-linguistic input comprises a prosodic input, and in accordance with a determination that the audio signal comprises the prosodic input, the action comprises a second action associated with the prosodic input.
4. The system of claim 3, wherein the second action is a modification of an action associated with a linguistic input.
5. The system of claim 3, wherein: the prosodic input is indicative of an emotion associated with the user of the wearable head device, and the action is further associated with the emotion.
6. The system of claim 1, wherein the method further comprises: determining whether the audio signal comprises a linguistic input; and in accordance with a determination that the audio signal comprises the linguistic input, performing a third action associated with the linguistic input.
7. The system of claim 6, wherein the action comprises a modification of the third action based on the non-linguistic input.
8. The system of claim 1, wherein the method further comprises: in accordance with a determination that the audio signal comprises the non-linguistic input, receiving information associated with the audio signal from a convolutional neural network, wherein the action is performed based on the information.
9. The system of claim 1, wherein the method further comprises: classifying a feature of the non-linguistic input, wherein the action is performed based on the classified feature.
10. The system of claim 1, wherein the method further comprises: associating the action with the non-linguistic input.
11. The system of claim 1, wherein the action comprises one of texting, performing an intent, and inserting an emoji.
12. The system of claim 1, further comprising: a sensor different from the microphone, wherein the method further comprises receiving, from the sensor, information associated with an environment of the system, wherein the action is further associated with the information received from the sensor.

13. The system of claim 1, wherein: the system comprises a mixed reality system in a mixed reality environment, the mixed reality system comprises the wearable head device, and the action is further associated with the mixed reality environment.

14. The system of claim 1, wherein: the method further comprises determining a position of the system, wherein the action is further associated with the position of the system.

15. The system of claim 1, wherein: in accordance with a determination that the system is associated with a first user, the action comprises a first action associated with the first user; and in accordance with a determination that the system is associated with a second user, different from the first user, the action comprises a second action associated with the second user, different from the first action.

16. The system of claim 1, wherein: the audio signal comprises a frequency-domain feature and a time-domain feature, and the determination of whether the audio comprises the non-linguistic input is based on the frequency-domain feature and the time-domain feature.

17. The system of claim 1, wherein the method further comprises: receiving information from a feature database, wherein the determination of whether the audio comprises the non-linguistic input is further based on the information.

18. The system of claim 1, wherein determining whether the audio signal comprises the non-linguistic input comprises using a first processor, and the method further comprises: in accordance with a determination that the audio signal comprises the non-linguistic input, waking up a second processor to perform the action.

19. The system of claim 1, wherein: the plurality of microphones comprises a first microphone and a second microphone, the first microphone is at a first location of the wearable head device, the first microphone configured to rest at a first distance from a user's mouth, and the second microphone is at a second location of the wearable head device, the second microphone configured to rest at a second distance from the user's mouth, the second distance unequal to the first distance.

20. A method comprising: receiving, using a plurality of microphones of a wearable head device, an audio signal; determining whether the audio signal is associated with a user of the wearable head device, wherein the determining whether the audio signal is associated with the user comprises: determining respective distances between a source of the audio signal and each of the plurality of microphones, and determining, based on the respective distances, a location of the source of the audio signal; in accordance with a determination that the audio signal is associated with the user of the wearable head device: determining whether the audio signal comprises a non-linguistic input; and in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input; and in accordance with a determination that the audio signal is not associated with the user of the wearable head device, forgoing determining whether the audio signal comprises the non-linguistic input.

21. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to execute a method comprising: receiving, using a plurality of microphones of a wearable head device, an audio signal; determining whether the audio signal is associated with a user of the wearable head device, wherein the determining whether the audio signal is associated with the user comprises: determining respective distances between a source of the audio signal and each of the plurality of microphones, and determining, based on the respective distances, a location of the source of the audio signal; in accordance with a determination that the audio signal is associated with the user of the wearable head device: determining whether the audio signal comprises a non-linguistic input; and in accordance with a determination that the audio signal comprises the non-linguistic input, performing an action associated with the non-linguistic input; and in accordance with a determination that the audio signal is not associated with the user of the wearable head device, forgoing determining whether the audio signal comprises the non-linguistic input.
